

NAYEL BENABDESADOK

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🌐 LinkedIn

🐙 GitHub

SUMMARY

Final-year dual-degree student at ENSAE Paris and École des Mines, seeking a graduate internship as a Quant Researcher, Algo Trader or Pricing Structurer in the commodities and weather markets for May-June 2027.

SKILLS

Coding Python, C#, R, SQL, C, Conda

Languages French (Native), English - Spanish (Fluent), Japanese (Beginner)

EXPERIENCE

- June – Dec 2025 **Quantitative Researcher Intern @ Natixis Corporate & Investment Banking** **Paris, France**
- Crafted a loan pricing model to monitor and mark-to-market Natixis' €100bn outstanding credit portfolio.
 - Engineered a regression-based methodology to proxy CDS curves for unquoted counterparties.
 - Implemented the full pricing pipeline and curve-adjustment modules in C# and integrated advanced regression metrics to enhance the reliability of the desk's decision-support tools.
- Mar – July 2026 **AI Engineer Intern @ Corpy&Co** **Tokyo, Japan**
- Deployed ML solutions for industrial clients, moving from initial prototyping to containerized production.
 - Architected a hybrid OCR-VLM extraction pipeline to digitize Japanese logistics delivery slips covering detection, semantic extraction, and geometric matching of structured table data.
- Dec - May 2025 **Research Assistant @ Milliman France** **Github - Paris, France**
- Explored a machine learning paradigm: Federated Learning applied to the insurance sector.
 - Implemented federated regression models for automobile risk prediction and evaluation of the results.
- June - Sept 2024 **Data Scientist Intern @ BPCE** **Paris, France**
- Developed Computer Vision algorithms for extracting unstructured PDF data (OCR, object detection), increasing file processing efficiency by 60 % and 30 minutes/day/manager.
 - Designed NLP models for optimising financial management (document clustering).
- 2022 - 2024 **Communications Officer @ Mines Études et Projets (University Consulting Firm)** **Saint-Étienne, France**
- Crafted fortnightly content, contributing to a turnover of €40k for the 2024 mandate.
 - Built a communication strategy (500 students) and management of social networks.

PROJECTS

- Mar 2026 - Now **Stochastic and ML Temperature Models for Weather Derivatives & Commodity Trading** **App Boreas**
- Ran an exploratory analysis of daily temperature profiles for US cities listed on the CME weather market.
 - Modelled temperature dynamics with an Ornstein-Uhlenbeck process and Machine Learning forecasters (Ridge, Random Forest, XGBoost), benchmarked on forecast accuracy and option pricing impact.
 - Researched weather-gas volatility coupling via Kalman-filter factor extraction and a GARCH-X model.
 - Deployed **Boreas**, an end-to-end interactive dashboard to price weather options and benchmark contract prices across temperature models.

EDUCATION

- 2024 - 2027 **ENSAE Paris - Institut Polytechnique de Paris** **MEng in Mathematical Statistics & Probability**
- Econometrics, Statistics, Time Series, Machine Learning and Data Science
 - Financial Mathematics, Stochastic Processes, Financial Instruments
- 2023 - 2024 **Seoul National University** **Computer Science Department**
- Computer Vision and Image Processing, Natural Language Processing (NLP), Deep Learning
- 2022 - 2024 **Mines Saint-Étienne** **MEng in Computer Science**
- Applied Mathematics, Data Science, Physics Modelling for Energy Transition
 - Bachelor's degree in Fundamental Mathematics (*Jean-Monnet University*)
- 2020 - 2022 **Collège Stanislas Paris** **"Classes Préparatoires"**
- Intensive program preparing for entrance to French engineering Grandes Écoles

OTHER

Sports : Swimming (6 years), Tennis (8 years), Badminton (1 year), Scuba Diving (PADI certification)

Trips : Algeria (Dual Nationality), Japan, South Korea, USA